

CS-UY 2214 — Homework 1

Adapted from Jeff Epstein, Modified by Ratan Dey

Complete the following exercises. Put your answers in a plain text file named `hw1.txt`. Number your solution to each question. When you finish, submit your file on Gradescope.

Please note that your solutions must be in a *plain text file*. Other formats, such as PDF, RTF, and Microsoft Word, will not be accepted. Here are some recommended editors that produce plain text files:

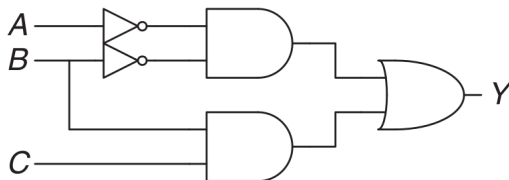
- Notepad (comes with Windows)
- TextEdit (comes with Mac OS); note that if you are using TextEdit, you need to select “Make Plain Text” from the Format menu before saving the file
- gedit (available on most Linux distributions)
- nano (available on most Linux distributions)
- Sublime Text
- VSCode
- Atom
- Vim
- Emacs

For questions that require a solution expressed as an image (specifically, question 3), submit the image as a separate file. The image file should be named `hw $n$ q $m$` , where n is the homework number and m is the question number; use an appropriate suffix (either `jpg` or `png`).

Please note that all homework must represent *solely your own work*. Any form of collaboration is strictly prohibited. Please see the syllabus for more details.

Problems

1. Consider the following circuit:



Complete the truth table for the circuit:

A	B	C	Y

- Give an algebraic expression for the circuit in the previous question.
- Draw a circuit diagram corresponding to the following expression:

$$Q = (A \& B) \mid ((B \mid C) \& (B \& C))$$

Do not simplify the equation. Submit your answer as an image, in accordance with the instructions at the beginning of this document. Your image may be a diagram created in a drawing program, or it may be a photograph of a hand-drawn diagram on paper.

- Consider the following C++ program. Do not run it.

```
#include <iostream>
using namespace std;
int main() {
    unsigned int x = 79;
    while (x <= 79)
        x--;
    cout << "x=" << x << endl;
    return 0;
}
```

Your friend Rufus argues that this program has an infinite loop, and will therefore never end. Is he right or wrong? Justify your opinion with a persuasive explanation, but do not type in or run the program. Predict the output of the program.

- Write the following decimal numbers in 8-bit 2's-complement binary. Write all eight bits, including leading zeros.
 - 52
 - 53
 - 77
 - 12
- Write the following 8-bit 2's-complement binary numbers in decimal.
 - 01111111
 - 10111111
 - 11011111
 - 01010101
- Assume that the following operations are calculated using 8-bit 2's complement numbers. Give the resulting sum in decimal. Take into account the effects of possible overflow.

- (a) $39 + 27$
 - (b) $99 + 30$
 - (c) $30 + -40$
 - (d) $-100 + -100$
8. Assume that the following operations are calculated using 8-bit unsigned numbers. Give the resulting sum in decimal. Take into account the effects of possible overflow.
- (a) $34 + 55$
 - (b) $99 + 30$
 - (c) $100 + 200$